MIL-H-43162F(GL)
26 July 1988
SUPERSEDING
MIL-H-43162E(GL)
13 May 1976

MILITARY SPECIFICATION

HAT, SERVICE, WOMAN'S, WOOL OR POLYESTER AND WOOL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This document covers a woman's wool or polyester and wool hat.

1.2 Classification. The hat shall be of one type and in the following sizes (see 6.2).

SCHEDULE OF SIZES

20 -1/2, 21, 21-1/2, 22, 22-1/2, 23, 23-1/2

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8410

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SPECIFICATIONS

FEDERAL

L-P-378	- Plastic Sheet and Strip, Thin Gauge, Polyolefin
V-T-276	- Thread, Cotton
JJ-W-155	- Webbing, Textile, (Cotton, Elastic)
DDD-L-20	- Label: For Clothing, Equipage, and Tentage, (General Use)
PPP-B-636	- Boxes, Shipping, Fiberboard
PPP-T-45	- Tape, Gummed, Paper, Reinforced and Plain, for Sealing and Securing

MILITARY

MIL-C-368	- Cloth, Satin, Rayon and Cloth, Twill, Rayon
MIL-C-823	- Cloth, Serge; Wool, Wool and Nylon, Polyester and Wool
MIL-L-35078	- Loads, Unit: Preparation of Semiperishable Subsistence Items; Clothing, Personal Equipment and Equipage, General Specification For
MIL-C-43920	- Cloth, Interlining, Cotton or Synthetic, and Nylon

STANDARDS

FEDERAL

FED-STD-191	- Textile Methods
FED-STD-751	- Stitches, Seams, and Stitchings

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-147	- Palletized Unit Loads
MIL-STD-731	- Quality of Wood Members for Containers and Pallets

(Copies of specifications, standards, and handbooks required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issues of the nongovernment documents which are current on the date of the solicitation.

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Guide sample. Samples, when furnished, are solely for guidance and information to the contractor (see 6.3). Variations from the document may appear in the sample, in which case the document shall govern.

3.2 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.2, 6.2, and 6.4).

3.3 Materials. (See 6.5).

3.3.1 Basic material. The basic material shall be Army Green 344 wool serge conforming to type I, class 8 of MIL-C-823 or shall be Army Green 344 polyester and wool serge conforming to type III, class 2 or class 3 of MIL-C-823.

3.3.2 Lining. The material for the lining shall be rayon cloth, shade Army Green 45, conforming to class 1 or class 2 of MIL-C-368.

3.3.3 Crown interlining. The crown interlining material shall be haircloth having an unbleached cotton warp and a natural horsehair filling and conforming to the requirements listed below when tested as specified in 4.3.1.1. As an alternate, cloth interlining conforming to type I or II of MIL-C-43920 may be used.

Weight, ounces per square yard	9.0 to 10.4
Yarns per inch, minimum:	
Warp	60, two-ply cotton yarn
Filling	90, (natural horsehair)

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Weave	2/8 (5 harness satin) (2 warp ends weaving as 1)
Stiffness, load pounds, minimum:	
Warp	0.070
Filling	0.090

3.3.4 Crinoline. The material for tip lining, band facing interlining, backing the interlining of the brim, and for extending the brim interlining shall be bleached cotton crinoline conforming to the following requirements when tested as specified in 4.3.1.1:

Weight, ounces per square yard	2.3 to 3.1
Breaking strength, pounds, minimum (total of warp and filling)	45
Stiffness, load pounds, minimum (total of warp and filling)	0.035
Yarns per inch, minimum:	
Warp	40
Filling	30
Weave	Plain

3.3.5 Felt. The material for interlining the brim shall be pressed felt, made from white, black, or natural fiber, conforming to the following requirements when tested as specified in 4.3.1.1:

Weight, ounces per square yard	9.0 to 10.5
Thickness, inches	0.065 to 0.075
Ball burst strength, pounds minimum	35
Starch and protein content including chloroform and water soluble material, percent, maximum	20
Stiffness load pounds, minimum (total of machine and cross directions):	
Before drycleaning	0.77
After drycleaning	0.65
Wool content	25 to 40 percent (balance to be viscose rayon)

3.3.6 Sweatband ribbon. The ribbon for sweatband shall be 7/8 to 1 inch wide black rayon or rayon and cotton grosgrain and shall show fastness to wet-drycleaning (surface type cleaning) and perspiration equal to or better than the standard sample or equal to or better than a rating of "good". The ribbon shall show fastness to crocking equal to or better than the standard sample or shall have an AATCC Chromatic Transference Scale rating of 3.5. Testing shall be as specified in 4.3.1.1, Table V.

3.3.7 Elastic webbing. The elastic webbing for attaching band to the hat shall be dyed black, conforming to type II, class 1 or class 2 of JJ-W-155, except that the requirements for colorfastness shall not apply.

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3.3.8 Thread. The thread shall be cotton, machine, type IA3, ticket No. 50, 3-ply and ticket No. 70, 2-ply for inside stitching, and type IC2, ticket No. 0, 3-ply for outside stitching, conforming to V-T-276. Type IC2 thread may be used for inside stitching. The thread shall be dyed Green E, C.A. 66034 except that the thread used for attaching the sweatband (operation 14.b, table II), shall be dyed Black AA, C.A. 66043. The black thread may be used as bobbin thread for operation 13.c, table II. Any of these threads may be used for basting.

3.3.8.1 Thread colorfastness. Both types IA3 and type IC2 threads shall show fastness to wet-dry cleaning equal to or better than the standard sample or equal to or better than a rating of "good". The type IC2 thread shall show fastness to light equal to or better than a rating of "good". Testing shall be as specified in 4.3.1.1, table V.

3.3.9 Label. Each hat shall conform to type VI, class 4 of DDD-L-20. The fastness to dry cleaning requirements of DDD-L-20 shall apply.

3.3.10 Tip protector. The material for tip protector and name card insert shall be polyethylene film, 0.004 inch thick, conforming to type I, class 1, grade A, B, or C, finish 1 of L-P-378.

3.4 Design. The hat, as shown on figure 1, is a fabric hat reinforced by interlinings and shaped over a millinery block. The hat has an oval shaped crown, an all-over stitched brim rolled at back and straight at front, and a removable band for insignia.

3.5 Figures. Figures are furnished for information only. To the extent of any inconsistencies between the written specification and the figures, the specification shall control.

3.6 Patterns. Sample Government blocks (one crown and one brim) will be loaned to the contractor so that he may procure a set of blocks to be used in the fabrication of the end item. Standard patterns for use in cutting working patterns will be furnished by the Government. Neither the Government patterns nor the contractor's working pattern shall be altered in any way. The seam allowance shall be 1/4 inch except as otherwise specified herein. Each hat, crown and brim, shall be properly blocked on the standard blocks to be furnished by the Government. The sample blocks shall be returned to the Government in good condition upon completion of contract or order.

3.6.1 Pattern parts. The component parts of the hat shall be cut from materials specified and in accordance with the number of pattern parts required for the manufacturing process as follows:

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TABLE I. List of pattern parts

Material	Pattern nomenclature	Cut parts
Basic material	Side crown 1/	1
	Tip 1/	1
	Brim	2
	Band	1
	Band facing 2/	1
Rayon lining	Side crown lining	1
	Tip lining 3/	1
Haircloth	Side crown 1/	1
	Tip 1/	1
or		
Alternate fabric	Side crown 5/	2
	Tip 5/	2
Crinoline	Band facing (interlining) 2/	1
	Tip lining 3/	1
	Brim interlining 4/	1
Felt	Brim interlining 4/	1
Polyethylene	Tip lining 3/	1
	Card insert	1
Shaper	Band	-

1/ The same pattern part is used for cutting the basic material and haircloth.

2/ The same pattern part is used for cutting the basic material and crinoline.

3/ The same pattern part is used for cutting the rayon, crinoline and polyethylene.

4/ The same pattern part is used for cutting the crinoline and felt.

5/ When alternate material is used for the crown interlining, see operation 1.a of table II.

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3.7 Construction.

3.7.1 Stitches, seams, and stitchings. Stitch, seam, and stitching types specified in table II, shall conform to FED-STD-751. When two or more seam or stitch types are given for the same part of an operation, any one of them may be used. Where stitch type 101 or 401 is used, the looper (underthread), shall be on the inside of the hat. Seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, or open seams result.

3.7.1.1 Stitches per inch. The minimum and maximum number of stitches per inch shall be as specified in table II.

3.7.1.2 Thread breaks and ends of seams. Ends of all seams and stitchings (except label), when not caught in other seams or stitching, shall be backtacked not less than 1/4 inch. Ends of a continuous line of stitching shall be overlapped not less than 1/2 inch. Thread breaks shall be secured by stitching back of the break not less than 1/2 inch. Skipped stitches or thread breaks on 401 or 101 seaming may be repaired using a 301 stitch type.

3.8 Manufacturing operations requirements. The hat shall be manufactured in accordance with operation requirements specified in table II. The contractor is not required to follow the exact numerical sequence of operations listed. Any additional basting or holding stitching to facilitate manufacture is permissible provided the thread is removed or does not show on the finished garment.

3.8.1 Shade marking. The component parts of the hat shall be shade marked or bundled to insure a uniform shade and size throughout the hat. Any method of shade marking may be used except:

a. Corrosive metal fastening devices. No metal device shall be used on rayon lining.

b. Sewn-on shade tickets.

c. Adhesive type tickets which leave an impression or traces of paper or adhesive on the material upon removal of the tickets.

3.8.2 Pressing and blocking. Pressing shall be accomplished with a heated steam pressing iron, except as otherwise specified in table II. Blocking shall be as specified in table II.

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TABLE II

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
1.	Cutting. a. Cut the hat in strict accordance with the patterns furnished, which show directional lines, size, and notches for proper assembly. Directional lines shall be placed in the warp direction of the material except that the directional line for the haircloth tip, when used, shall be in the filling direction. When alternate crown interlining is used, two sets of parts shall be cut, one with directional line in warp direction and the other with directional line in the filling direction. b. Cut the parts from the basic material as indicated on figure 2, which shows the positioning of the pattern parts in relation to the warp direction of the material and the twill lines in the finished hat. Twill lines in finished hat may vary from requirement by not more than 3/16 inch. c. Cut the felt in any direction. d. Cut crinoline stripping (for extending edge of brim interlining) on the bias 1 inch wide. e. Cut the elastic webbing for hatband sufficiently long so that it will not constrict the hat when band is in place. The length of the elastic shall provide for 3/8 to 1/2 inch extension on both ends for overlapping ends on band (see operation 7.g)						

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
2.	Replacement of damaged parts. Care shall be exercised during the spreading, cutting and manufacturing operations to assure that material defects and damages, as classified in table VI, are excluded and replaced with non-defective and properly matched material. NOTE: Unit containing one or more clearly visible mends will not be acceptable.						
3.	Shade markings. Mark, ticket or bundle all parts to insure a uniform shade and proper assembly of the hat (see 3.8.1).						
4.	Make crown interlining. Finished appearance. When haircloth interlining is used, the crown interlining shall be fabricated such that the light side will be against the basic material shell in the finished hat. When alternate interlining material is used, two separate complete crown interlinings shall be assembled with one fabricated from material cut in warp direction and the other from material cut in the filling direction.						
	a. Join back seam of side crown piece by overlapping the edge 1/2 inch and stitching 1/4 to 5/16 inch from edge.	101 or 301 or 401	LSa-1 LSa-1 LSa-1	10-14 10-14 10-14	50-3 50-3 50-3	--- 50-3 70-2	

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
4.	Make crown interlining. (cont'd)						
	b. Place one or two rows of tension stitches around tip, 1/8 inch gage with the outside row 1/8 to 3/16 inch from edge, to cup the tip and to provide sufficient resiliency.	301	OSa-2	10-14	50-3	50-3	
	NOTE: This operation may be omitted providing that the tip on the finished hat will recover its cup shape after moderate pressure is applied to the center of the tip and then removed.						
	c. Join tip to side crown by overlapping the tip 1/2 inch on outside of side crown, matching center fronts and center backs, and stitching 3/16 to 1/4 inch from edge.	101 or 301 or 401	LSa-1 LSa-1 LSa-1	10-14 10-14 10-14	50-3 50-3 50-3	--- 50-3 70-2	
	d. Place a second row of stitching 1/16 to 1/8 inch above the first row of stitching (see operation 4.c).	101 or 301 or 401	LSa-1 LSa-1 LSa-1	10-14 10-14 10-14	50-3 50-3 50-3	--- 50-3 70-2	
5.	Make crown.						
	Finished appearance. The stitching on the joining seams shall be unwavering and the top stitching shall be uniform. The stitch tension shall be properly adjusted so that the thread of the joining seam will not be exposed on the finished blocked hat.						
	a. Join back seam of side crown piece.	301	SSz-3(a)	10-14	50-3	50-3	

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
5.	Make crown. (cont'd)						
	b. Stitch tip to side crown with a 1/4 inch seam allowance, matching center fronts and center backs with the back seam of the side crown spread open.	301	SSz-3(a)	10-14	50-3	50-3	
	c. Spread back seam and crown seam open and top stitch with each line of stitching 1/16 to 3/32 inch from joining seam. If a double needle machine is used, the gage shall be 1/8 to 3/16 inch and the seam line shall be centered between the rows of stitching. The seams may be pressed open prior to top stitching.	301	SSz-3(b)	10-14	0	0	
	NOTE: The back seam may be top stitched prior to stitching tip to side crown.						
6.	Make brim.						
	Finished appearance. The brim shall consist of two plies of the basic material interlined with a ply of felt and a ply of crinoline. The felt ply shall be on top of the crinoline in the finished brim and shall be next to the basic material on the top side of the brim at center front of finished hat. The outer edge of the brim shall be smooth with the joining seam fully and evenly worked out. The brim shall be smooth, without wrinkles, pleats or fullness between rows of stitching.						

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make brim. (cont'd)						
	a. Place felt interlining over crinoline interlining and join center back seam with the stitching on the felt side. (The felt and crinoline may be basted together prior to joining back seam.)	101 or 301 or 401	SSa-1 SSa-1 SSa-1	10-14 10-14 10-14	50-3 50-3 50-3	--- 50-3 70-2	
	b. Spread seam open and press firmly with a heated steam iron.						
	c. Overlap the bias strip 1/2 inch (see operation 1.d) on the outer edge (felt side) of the interlining and stitch to brim interlining 1/8 to 3/16 inch from inner edge of strip through all plies. Overlap the ends of the strip 1/4 to 3/8 inch, at back of brim, 3/8 to 1/2 inch to the right or left of back seam.	101 or 301 or 401	LSa-1 LSa-1 LSa-1	10-14 10-14 10-14	50-3 50-3 50-3	--- 50-3 70-2	
	d. Join center back seam of each brim piece (basic material).	301	SSa-1	10-14	50-3	50-3	
	e. Press seams open.						
	f. Superimpose the two basic material brim pieces with the fabric face to face with seam on seam. Baste the pieces together 1/8 to 1/4 inch from outer edge.	101	SSa-1	6-8	50-3	---	
	g. Place brim (top and underbrim) on the felt side of brim interlining with the top brim against the felt and with the outer edge even with the outer edge of the crinoline strip and baste stitch 1/8 to 3/16 inch from outer edge.	101	SSa-1	6-8	50-3	---	

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					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make brim. (cont'd)						
	h. Place row of stitching 1/8 inch outside the edge of interlining through top brim, underbrim, and crinoline strip.	301	SSa-1	10-14	50-3	50-3	
	- or -						
	i. Place top brim, face up, on the felt side of brim interlining with the outer edge of brim even with outer edge of crinoline strip and baste stitch 1/8 to 3/16 inch from outer edge.	101	SSa-1	6-8	50-3	---	
	j. Place underbrim over top brim, face to face, with outer edges even and back seam superimposed and stitch 1/8 inch from outside edge of interlining, through crinoline strip, top brim and underbrim.	301	SSa-1	10-14	50-3	50-3	
	k. Trim excess seam allowance of basic material and crinoline strip around outer edge to 1/4 inch from brim joining seam. (This is approximately 1/8 inch from brim-interlining layers.)						
	l. Turn brim, work out seam and baste edge of brim. Smooth out all plies and baste the remainder of brim with four to six rows of stitching 1/2 to 5/8 inch apart.	101 101	SSae-2(b) and SSv-3	2-6 2-6	50-3 50-3	--- ---	
	m. Baste inner raw edges of brim together through all plies, 1/4 to 3/8 inch from edge.	101	SSa-1	6-8	50-3	---	

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					NEEDLE	BOBBIN/ LOOPER	COVER
6.	Make brim. (cont'd)						
	n. Stitch outer edge of brim (top side up) 1/16 to 3/32 inch from edge around the entire page.	301	SSae-2(b) and	10-14	0	0	
	Place a second separate row 1/16 to 1/8 inch from the first and commence stitching spirally the entire brim with rows of stitching 1/16 to 1/8 inch apart, removing basting stitches one row at a time ahead of spiral quilting stitching.	301	SSv	10-14	0	0	
	o. Mark or notch inside brim edge at center front (directly opposite center back seam). Slash around inner edge at intervals to permit proper fit of brim to inside of block (see operation 12.c).						
7.	Make band.						
	Finished appearance. Top edge of band shall be uniformly stitched and the band shall finish smooth and flat. The front of the finished band shall conform to the band shaper (3.6.1) (and the crinoline interlining shall extend to the bottom of finished band).						
	a. Superimpose crinoline band facing interlining on basic material band facing and baste 1/8 to 3/16 from bottom edge. In addition, the two plies may be basted together along the length of the band (thread to be removed later).	101	SSa-1 and OSf-1	4-6	50-3	---	

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					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make band. (cont'd)						
	b. Superimpose basted band facing (with interlining face up) on basic material band with all top and side edges even. Stitch top and side edges 1/4 inch from edge, maintaining contour of the edges.	301	SSe-3(a)	10-14	50-3	50-3	
	c. Trim seam allowance and V-notch out each corner of upper rounded area to within 1/16 inch of the stitch line (width of V-notch at top shall be 1/4 to 3/8 inch wide). Turn, work out edges evenly, and baste around top, sides and bottom edges 1/8 to 1/4 inch from edge. The inside corners at ends of curved portion shall conform to the inside corners of band shaper.	101	SSe-1(b)	4-6	50-3	---	
	d. Stitch top of band 1/16 to 3/32 inch from edge; place two additional rows of stitching around top of band with spacing between each of the three rows 1/16 to 1/8 inch gage.	301	SSe-3(b)	10-14	0	0	
	e. Press partially finished band and check band with finished shaper to assure that band is in conformity with required shape.						
	f. Overedge bottom edge of band. (This operation may be omitted.)	501 or 502 or 503	SSa-1 SSa-1 SSa-1	8-10 8-10 8-10	50-3 50-3 50-3	--- 50-3 50-3	

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NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
7.	Make band. (cont'd)						
	g. Fold bottom edge of band firmly over raw edge of interlining and facing, catching the ends of the elastic under the folded edge at each end of band (see operation 1.e). Hand or machine fell or catch stitch (crisscross) raw edge to the band facing. (Care shall be taken not to stretch bottom edge of band. Band shall not be pressed after felling, and stitching impressions or stitching shall not show on outside of band.)	Hand or 204 103	--- --- ---	4-6 4-6 4-6	50-3 50-3 50-3	--- --- ---	
	h. Securely tack or bartack ends of elastic through all plies across the width of the elastic.	Hand or machine	---	4-6 21-28 per bartack	50-3 50-3	--- 50-3	
8.	Make lining. Finished appearance. The overlap of the polyethylene film for name card insert shall be horizontal to front and the double ply shall be at front in the finished hat. The pleats shall be equally distributed and shall face the same direction. a. Superimpose tip lining pieces as follows: (1) Crinoline (2) Rayon (3) Polyethylene film tip (4) Polyethylene film name card insert						

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					NEEDLE	BOBBIN/ LOOPER	COVER
8.	Make lining. (cont'd)						
	b. Stitch plies of tip together 1/4 to 3/8 inch from edge.	101 or 301 or 401	SSa-1 SSa-1 SSa-1	6-8 6-8 6-8	50-3 50-3 50-3	--- 50-3 70-2	
	c. Stitch side crown lining to tip distributing fullness with pleats, 1/8 to 1/4 inch in depth, turning in raw edge of back seam allowance into a pleat. Pleats may be stitched down prior to setting lining to tip.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-3 50-3	50-3 70-2	
	d. The back joining seam of the lining shall be left open for adjusting to the various sizes of hats (see operation 11.b).						
	-or-						
	e. The back seam may be joined and the excess fullness distributed evenly along the bottom edge of the crown when stitching lining to crown at operation 11.a.	301 or 401	SSa-1 SSa-1	10-14 10-14	50-3 50-3	50-3 70-2	
9.	Block crown.						
	Finished appearance. The crown (basic material and interlining) shall fit together smoothly and evenly without twists, pleats, puckers, wrinkles, creases or fullness. The basic material top crown and side crown seams shall be centered over their corresponding interlining seams. The finished crown shall be free of gloss, shine or excessive seam impression marks and shall not be distorted, crushed or dented.						

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NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Block crown. (cont'd) a. Haircloth interlining (when used). 1. Place crown interlining (light side of haircloth outside) on correspondingly sized, pre-heated block (with the wood suitably covered). Match center front, center back and top joining seam to the appropriate block markings and scored top crown line. 2. Cover crown interlining with pressing cloth, (see 6.6) securing cloth to crown with tie cord securely fastened around block base. Wet the pressing cloth and pull down firmly on block. 3. Steam press entire crown thoroughly. Remove cord and pressing cloth. b. Alternate material interlining (when used). 1. The crown cut in filling direction shall be blocked first as in operation 9.a. 2. The crown cut in the warp direction shall be placed over the crown cut in the filling direction with the back seams of the crowns off center by not less than 1/2 inch or more than 1 inch. 3. The two crowns together shall be blocked as in operation 9.a.						

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
9.	Block crown. (cont'd)						
	c. Place basic material crown on right side over pre-heated, correspondingly sized crown interlining and block, centering seams on seams and matching center front and center back. Place pressing cloth over assembly and securely fasten tie cord around base at indentation. Steam to eliminate wrinkles, puckers, tightness, fullness, gloss or shine. Check height measurements. Remove cloth.						
	d. Place assembly in heated oven until thoroughly dry. Remove cord when dry.						
	NOTE: Pins may be used to help secure crown interlining or basic material crown during the blocking operations.						
10.	Finish crown.						
	a. Check mark on center front of crown assembly and re-mark, if necessary (directly opposite center back seam); remove crown assembly from block.						
	b. Stitch plies together at cord mark. Place a second row 1/16 inch above the first row.	101 or 301 or 401	SSa-1 SSa-1 SSa-1	10-14 10-14 10-14	50-3 50-3 50-3	--- 50-3 70-2	
	c. Place a row of baste stitching (not to be removed) through all plies 3/8 to 1/2 inch above bottom row of stitching in operation 10.b.	101 or Hand or 301	SSa-1 SSa-1 SSa-1	4-14 4-14 4-14	50-3 50-3 50-3	--- --- 50-3	

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
10.	Finish crown. (cont'd)						
	d. Measure front side and back height of crown to conform with measurements and mark crown accordingly. Trim excess material around crown along the marked line.						
11.	Attach lining.						
	a. Position lining into crown with lining tip and crown tip aligned and stitch 1/4 to 5/16 inch from bottom edge of basic material crown, catching folded edge of lining at back. If back seam has been joined (see operation 8.e) distribute the fullness evenly around bottom edge of crown.	101 or 301 or 401	SSa-1 SSa-1 SSa-1	10-14 10-14 10-14	50-3 50-3 50-3	--- 50-3 70-2	
	b. Hand fell back seam of lining if seam was left open. Stitching shall not catch crown interlining.	Hand	---	6-8	50-3	---	
	c. Trim lining even with bottom edge of crown.						
12.	Block brim.						
	Finished appearance. The brim shall finish smooth without wrinkles, creases, puckers, gloss marks or shine. Stiffness after blocking shall meet or exceed stiffness before blocking. Contours, shape and dimensions shall conform to appropriately sized block. Back and side edges of the outside brim shall be aligned with the top edge of block groove and front of brim shall be aligned with front block edge. Center front, sides and back brim widths shall finish in accordance with the blocks and the table of measurements.						

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Block brim. (cont'd) a. Steam brim block quickly in a steam box and wipe dry. b. Place block in heated oven until hot. c. Slash inner edge of brim, if additional slashes to those made in operation 6.o are required to assure a firm fit of brim to inside of block. d. Dampen inside edge of brim with wet sponge and steam press to soften brim at inside. Place brim in steam box and steam quickly. e. Place steamed brim on pre-heated brim block, with top brim (basic material felt) on outside of block at sides and back and at inside of block along front brim. Match center back brim seam to block and pin to block in at least two places, with brim edge even with top edge of groove. Place assembly in steam box and steam quickly. f. Start pinning at center back of brim and continue pinning around sides. Alternate pinning and steam pressing of brim, shaping it to fit outside of block with brim edges even with top edge of groove. Match center front of brim to center front of block and with brim edges aligned with front block edge, continue pinning, steam pressing and shaping, until brim is pin-fitted to block. Place assembly in steam box and steam quickly.						

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
12.	Block brim. (cont'd) g. Smooth remainder of brim over the block, rolling it to inside of the block. Alternately, pin, steam press and shape until entire brim is in place on the block. Move pins and set plunger (plug), locking inner brim into position. Place entire assembly in steam box and steam quickly. Remove from box and dry with hot air hose. h. Remove pins one at a time, pressing with a steam iron during the removal process, to remove pin marks and shine. Dampen entire brim slightly with wet sponge. Cover assembly with a pressing cloth (see 6.6) and fasten cloth to the block with a blocking cord secured around the block in the outside groove. i. Press covered assembly with steam iron to shrink-fit it to the block. Place entire assembly in heated oven until brim is thoroughly dry. Remove cord, press cloth, plunger and brim from block. j. Trim excess material along inside brim edge to within 3/8 to 1/2 inch of the band line. Slash seam allowance at intervals along inner edge with depth of cut nor more than 1/4 inch.						
13.	Attach brim. Finished appearance. The brim shall fit the crown smoothly and evenly with the joining seam smooth and flat both inside and outside. The center back crown and brim seam shall be aligned and the center front of the brim shall be aligned to the center front of the crown. Front of brim shall be smooth, even and symmetrical (reflecting the shape of the block).						

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NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
13.	Attach brim. (cont'd)						
	Both turned up brim sides shall be smooth, even, symmetrical and equidistant from crown. Finished brim widths at center front, center back and center sides shall conform to table of measurements and the appropriately sized block dimensions.						
	a. Mark center front of brim on outside, turn brim down and mark inside edge at center front, center sides, and center back, connecting the marks with a continuous line to identify the brim joining seam.						
	b. Position brim inside crown, matching center fronts and center backs and tack stitch together 1/8 to 3/16 inch from edge.	Tack stitch	---	---	50-3	---	
	c. Stitch crown to brim 1/8 to 1/4 inch from crown edge. If desired to use this as a guideline stitch for operation 14.b, black thread (see 3.3.8) may be used for the bobbin. (Black thread shall not be visible on finished hat).	301	LSa-1	10-14	50-3	50-3	
14.	Attach sweatband and label.						
	Finished appearance. The label shall be positioned 1 to 2 inches to the left of the center back in the finished hat (as worn). The bottom of label shall be at the top of the sweatband. The free edge of the sweatband shall not be tight and shall fit the crown smoothly and evenly.						

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TABLE II (cont'd)

NO.	MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
14.	Attach sweatband and label. (cont'd)						
	a. Lap the edge of the sweatband ribbon over the bottom margin of label and stitch 1/16 to 1/8 inch from edge of ribbon. The stitching shall extend not less than 1/2 inch beyond each end of label.	301	LSa-1	10-14	50-3	50-3	
	b. Position the sweatband ribbon to the inside of hat with the bottom edge covering the joining-seam stitching with one end turned in not less than 3/8 inch, overlapping the other end at center back. Hand fell or machine stitch around bottom edge. Securely tack folded edge at center back.	Hand or 301	--- ---	6-8 10-14	50-3 50-3	--- 50-3	
15.	Position band.						
	Position band on hat with center of band aligned with center of hat and with bottom edge of band covering raw edge of crown. The band when in position shall not constrict the hat.						
16.	Clean hat.						
	Trim thread ends and remove loose threads inside and outside; remove all spots, stains and shade tickets.						
17.	Size hat and prepare for packing.						
	a. Dampen sweatband slightly and place on an appropriately sized solid metal heated sizing block with cloth between hat and block to protect polyethylene crown.						

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NO.	TABLE II (cont'd) MANUFACTURING OPERATIONS REQUIREMENTS	STITCH TYPE	SEAM AND STITCHING TYPE	STITCHES PER INCH	THREAD		
					NEEDLE	BOBBIN/ LOOPER	COVER
17.	Size hat and prepare for packing. (cont'd) b. Touch up with steam pressing iron to remove gloss or wrinkles due to handling. Allow to dry thoroughly. c. Prepare hats for packing (see section 5).						

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3.9 Measurements. The finished hats shall conform to the measurements shown in table III.

TABLE III. Measurements in inches

Size	Crown height at back seam	Crown height at center front	Brim depth at center front	Brim height at center back	Brim height at center sides	Band height at center front
20-1/2	2-13/16	2-3/4	2-3/16	2-7/16	1-7/16	2-5/8
21	2-13/16	2-3/4	2-3/16	2-7/16	1-7/16	2-5/8
21-1/2	2-7/8	2-7/8	2-1/4	2-1/2	1-1/2	2-5/8
22	2-7/8	2-7/8	2-5/16	2-1/2	1-1/2	2-5/8
22-1/2	2-7/8	2-7/8	2-3/8	2-1/2	1-1/2	2-5/8
23	3-1/8	3	2-7/16	2-9/16	1-9/16	2-5/8
23-1/2	3-1/8	3	2-7/16	2-9/16	1-9/16	2-5/8
Tolerance	+ 1/4, - 1/8	± 1/8	+ 1/4, - 1/8	± 1/8	± 1/8	± 1/8

3.9.1 Methods of measuring. The hat shall be measured as follows:

Size. Place hat on corresponding size crown block. Correctly sized, the bottom of the hat shall be between 1 to 1-1/2 inches above the indentation line of block.

Crown height at back seam. Measure center back seam from top crown seam to bottom edge.

Crown height at center front. Center front is directly opposite center back seam. Measure center front from top crown seam to bottom edge.

Brim depth at center front. Measure from brim-to-crown joining seam at center front, to outside edge of top of brim.

Brim height at center back. Measure outside along center back seam, from top edge to turned bottom.

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Brim height at center sides. Measure at both sides halfway between back and front from top edge to turned bottom.

3.10 Workmanship. The hat shall conform to the quality of product established by this document and the occurrence of defects shall not exceed the applicable acceptable point values.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this document where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this document shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the document shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for assuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to assure compliance with all dimensional requirements.

4.1.3 Certificates of compliance. When certificates of compliance are submitted, the Government reserves the right to inspect such items to determine the validity of the certification.

4.2 First article inspection. When a first article is required (see 3.2), it shall be examined for the visual defects specified in 4.3.3 and for dimensions specified in 4.3.4.

4.3 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.3.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this document or applicable purchase document.

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4.3.1.1 Component and material testing. In addition to testing provisions contained in subsidiary specifications, testing shall be performed on components listed in table V for characteristics noted. Wherever applicable, tests shall be conducted in accordance with methods prescribed in FED-STD-191. When the data in the "Determination per sample unit" and the "Results reported as" columns are not specified in the table they shall be as required by the referenced test method. All test reports shall contain the individual values utilized in expressing the final result. Unless otherwise specified, certificates of compliance will be considered acceptable as evidence that the material characteristics for all tests in 4.3.1.1 and table IV and table V conform to the specified requirements when tested by the specified methods. The lot shall be unacceptable if one or more sample units fail to meet test requirements specified. Sampling for testing shall be in accordance with table IV. The unit for expressing lot size and the sample unit for testing shall be as follows:

Component	Lot size expressed as	Sample unit for testing
Haircloth	Yards	1-1/4 yards full width
Crinoline	Yards	1/4 yard full width
Felt	Yards	1 yard full width
Ribbon	Yards	2 yards

TABLE IV. Sample size

Lot size (lot unit)	Sample size
800 or less	2
801 to 22,000	3
22,001 and up	5

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TABLE V. Component tests

Component	Characteristic	Requirement reference	Test method	Determinations per sample unit	Results reported as
Haircloth	Material identification	3.3.3			
	Warp		1200	1	Pass or fail 1/
	Filling		2100	1	Pass or fail 1/
	Weight	3.3.3	5041	-	Average to nearest 0.1 ounce
	Yarns per inch:	3.3.3	5050	-	-
	Yarn ply	3.3.3	Visual	1	Pass or fail
	Weave	3.3.3	Visual	1	Pass or fail
	Stiffness:	3.3.3	5202	-	Average to nearest 0.01 load lb.
Crinoline	Material identification:	3.3.4	1200	1	Pass or fail 1/
	Weight	3.3.4	5041	-	Average to nearest 0.1 ounce
	Breaking strength	3.3.4	5100	-	-
	Yarns per inch:	3.3.4	5050	-	-
	Weave	3.3.4	Visual	1	Pass or fail
	Stiffness:	3.3.4	5202	-	Average to nearest 0.001 load lb.

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TABLE V. Component tests (cont'd)

Component	Characteristic	Requirement reference	Test method	Determinations per sample unit	Results reported as
Felt	Weight	3.3.5	5041	-	Average to nearest 0.1 ounce
	Thickness	3.3.5	5030	-	-
	Ball burst strength	3.3.5	5120	-	-
	Nonfibrous material	3.3.5	2611	-	Average to nearest percent
	Stiffness:				
	Before drycleaning	3.3.5	5202	5	Average to nearest 0.01 load lb.
	After drycleaning	3.3.5	5202 2/	5	Average to nearest 0.01 load lb.
	Fiber content	3.3.5	2100	-	Average to nearest percent
Ribbon	Material identification	3.3.6	1500, 1510, or 1200 (as applicable)	-	Pass or fail 1/
	Colorfastness:				
	Wet-drycleaning	3.3.6	5622	-	-

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TABLE V. Component tests (cont'd)

Component	Characteristic	Requirement reference	Test method	Determinations per sample unit	Results reported as
Ribbon (cont'd)	Perspiration	3.3.6	5680	-	-
	Width	3.3.6	5020	-	-

1/ A certificate of compliance shall be acceptable for this characteristic.

2/ The specimens shall be subjected to drycleaning in accordance with drycleaning procedure outlined in Method 5622 of FED-STD-191.

4.3.2 In-process inspection. Inspection of sub-assemblies shall be made to ascertain that construction details which cannot be examined in the finished product are in accordance with specified requirements. When examining the alternate interlining crown assembly, any failure to comply with requirements of operation 9.c of table II shall result in rejection of the crown assembly being inspected. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated non-conformance.

4.3.3 End item visual examination. The end items shall be examined for the defects listed in table VI and the assessment of penalty points shall be as indicated therein. Sampling and acceptance criteria shall be as specified in 4.3.3.1.

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TABLE VI. End item visual defects

Examine	Defect	Point value
Blocking and pressing:	a. Blocking or pressing omitted or any part of hat distorted, crushed, misshaped, twisted, stretched, excessively full or dented	3
	b. Gloss marks; shine, or seam impressions	2
Material defects and damages:	a. Any hole, cut, tear, needle chew, burn, mend, darn, patch, multiple float, loose or unsightly slub or shade bar	
	1. On outside crown or brim	3
	2. On inside	1
	b. Permanent crease in cloth	2
Shaded parts:	a. Outside part shaded	3
Cleanness:	a. Any spot or stain, clearly noticeable	
	1. On outside	3
	2. On inside	1
	b. One or more marking tickets not removed	1
	c. Any shade marking visible on outside	2
	d. Thread ends not trimmed, or loose threads not removed	1
	e. Visible basting thread	1

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Point value
Component and assembly:		
Cutting:	a. Any component part or required operation omitted (unless otherwise classified herein)	3
	b. Any operation not performed as specified (unless otherwise classified herein)	2
Seams and stitching:	Any component part not cut in conformance with directional lines indicated on patterns or not cut in accordance with specification requirements	3
	a. Accuracy of seaming:	
	1. Seam or component part twisted, puckered or pleated	3
	2. Part of material caught in unrelated operation or stitching:	
	- up to 1/4 inch inclusive	1
	- more than 1/4 inch	2
	3. Ends of stitching, when not caught in other seams or stitchings or when continuous line not overlapped, back-tacked less than 1/4 inch	2
	4. Stitching overlapping ends of thread break less than 1/2 inch	2
	b. Gage and stitching and seam allowance:	
	1. Irregular, i.e., unevenly gaged or corresponding stitchings not uniform (to be scored only when condition exists along major portion of seam)	2

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Point value
Seams and stitching: (cont'd)	2. Not width specified or not within range of gage specified, or varies more than 1/16 inch when no range is specified	1
	c. Stitches broken or missing:	
	1. Up to 1/4 inch inclusive	1
	2. More than 1/4 inch to 1/2 inch inclusive	2
	3. More than 1/2 inch	3
	d. Skipped stitches:	
	1. 1/4 inch to 1/2 inch long, inclusive	2
	2. More than 1/2 inch long	3
	e. Runoffs:	
	1. 1/8 inch to 1/4 inch long inclusive	1
	2. More than 1/4 inch long	2
	f. Raw edge:	
	1. On outside	2
	2. On inside, more than 1/8 inch long	1
	g. Seam and stitch type not as specified	2
	h. Stitches per inch more than maximum or less than minimum specified	1

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Point value
Seams and stitching: (cont'd)	i. Stitch tension:	
	1. Loose tension resulting in a loose seam	3
	2. Loose tension (exposed loops of top or bottom threads):	
	a. 1/4 to 1/2 inch in length	1
	b. More than 1/2 inch	2
	3. Tight tension (stitches break when strain is applied in the direction of the seam or stitching)	3
Labels:	a. Missing, incorrect or illegible	3
	b. Not securely caught in the stitching	1
	c. Stitching beyond end of label less than 1/2 inch	1
	d. Not positioned as specified	1
	e. The stitching through the printing	1
Crown:	a. Excessive fullness in crown interlining	3
	b. Fullness on either side of top stitching at tip	2
	c. Twill line at front of crown more than 3/16 inch from perpendicular	3
	d. Crown off center at front by more than 3/16 inch	3

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Point value
Band:	a. Less than three rows of stitching at top edge	2
	b. Elastic not proper length causing band to be loose or tight	2
	c. Elastic not inserted under fold at bottom of band or not securely tacked to band	3
	d. Twill line on front of band more than 3/16 inch from the perpendicular	3
	e. Crinoline not extending to bottom folded edge of band	2
	f. Felling or blindstitching visible on outside of band	2
	g. Bottom of band pressed, resulting in a creased bottom edge	2
Brim:	a. Fullness between rows of spiral stitching	2
	b. Edge of brim irregular, or not properly worked out	2
	c. Back seam crooked by more than a total variance of 1/8 inch from a straight line	1
	d. Back seam of brim not aligned with crown back seam by more than 1/4 inch	2
	e. Variance of more than 1/4 inch of space between side crown and brim on right and left sides, measured at points equidistant from back seam of brim	3

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TABLE VI. End item visual defects (cont'd)

Examine	Defect	Point value
Brim: (cont'd)	f. Brim off center at front by more than 3/16 inch	3
	g. Front edge of brim not smooth, not even, not symmetrical (not reflecting the shape of the block)	3
Lining:	a. Tight, short, twisted, or loose	3
	b. Pleats not evenly distributed	1
	c. Pleats less than 1/8 inch or more than 1/4 inch deep	1
	d. All pleats not facing in the same direction	1
	e. Card insert of crown protector piece not horizontal within 1/2 inch to front of hat or not front part of tip	1
Sweatband:	a. Exposed beyond bottom edge of hat	2
	b. Back end turned under less than 3/8 inch	1
	c. Back folded end not tacked securely	1
	d. Black thread (when used to attach brim) exposed below sweatband	1

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4.3.3.1 Sampling and acceptance for visual examination. The sampling and acceptance criteria shall be as specified in table VII and 4.3.3.1.1 through 4.2.3.1.3.4. The lot size unit and the sample unit shall be one hat. The lot shall be unacceptable if:

a. The point value for 3 and 2 point defects exceeds the applicable maximum acceptable point value.

b. The point value for total (3, 2 and 1 point) defects exceeds the applicable maximum acceptable point value.

4.3.3.1.1 Initiation of inspection. Normal inspection shall be used at the start of inspection unless otherwise directed by the responsible activity.

4.3.3.1.2 Continuation of inspection. Normal, tightened or reduced inspection shall continue unchanged on successive lots except where stitching procedures in 4.3.3.1.3 require change.

4.3.3.1.3 Switching procedures.

4.3.3.1.3.1 Normal to tightened. When normal inspection is in effect, tightened inspection shall be instituted when two out of five consecutive lots have been rejected on original inspection (i.e., ignoring resubmitted lots for this procedure).

4.3.3.1.3.2 Tightened to normal. When tightened inspection is in effect, normal inspection shall be instituted when five consecutive lots have been considered acceptable on original inspection.

4.3.3.1.3.3 Normal to reduced. When normal inspection is in effect, reduced inspection shall be instituted providing that all of the following conditions are satisfied:

a. The preceding ten lots have been on normal inspection and none have been rejected on original inspection; and

b. The total number of points for 3 and 2 point defects in the samples from the preceding ten lots is equal to or less than 60 percent of the total maximum acceptable point values for 3 and 2 point defects from the preceding ten lots; and

c. The total number of points for 3, 2 and 1 point defects in the sample from the preceding ten lots is equal to or less than 75 percent of the total maximum acceptable point values for 3, 2 and 1 point defects from the preceding ten lots; and

d. Production is at a steady rate; and

e. Reduced inspection is considered desirable by the procuring agency.

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4.3.3.1.4 Reduced to normal. When reduced inspection is in effect, normal inspection shall be instituted if any of the following occur on original inspection:

a. A lot is rejected; or

b. A lot is considered acceptable but exceeds the applicable first value (see 1/ to table VII); or

c. Production becomes irregular or delayed; or

d. Other conditions warrant that normal inspection be instituted.

TABLE VII. Sampling and acceptance criteria for visual examination

	Lot size	Sample size	Maximum acceptable point value: 3 & 2 point defects	Maximum acceptable point value: 3, 2 & 1 point defects
Normal inspection	Up thru 90	20	3 points	10 points
	91 thru 150	32	5 points	13 points
	151 thru 280	50	8 points	16 points
	281 thru 500	80	14 points	24 points
	501 thru 1200	125	19 points	34 points
	1201 thru 3200	200	27 points	48 points
	3201 thru 10,000	315	38 points	72 points
	10,001 thru 35,000	500	57 points	108 points
	35,001 and over	800	81 points	164 points

	Lot size	Sample size	Maximum acceptable point value: 3 & 2 point defects	Maximum acceptable point value: 3, 2 & 1 point defects
Tightened inspection	Up thru 150	32	3 points	10 points
	151 thru 280	50	5 points	13 points
	281 thru 500	80	8 points	16 points
	501 thru 1200	125	14 points	25 points
	1201 thru 3200	200	22 points	40 points
	3201 thru 10,000	315	32 points	60 points
	10,000 thru 35,000	500	49 points	90 points
	35,000 and over	800	73 points	136 points

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TABLE VII. Sampling and acceptance criteria for visual examination (cont'd)

	Lot size	Sample size	Maximum acceptable point value: 3 & 2 point defects	Maximum acceptable point value: 3, 2 & 1 point defects
Reduced inspection	Up thru 280	20	3-11 points	9 points
	281 thru 500	32	5-14 points	15 points
	501 thru 1200	50	8-16 points	21 points
	1201 thru 3200	80	14-22 points	30 points
	3201 thru 10,000	125	19-27 points	40 points
	10,001 thru 35,000	200	27-35 points	60 points
	35,001 and over	315	35-44 points	90 points

1/ If the first value is exceeded but the second value has not been exceeded, the lot but reinstate normal inspection (see 4.3.3.1.4b). The second value is the maximum acceptable point value.

4.3.4 End item dimensional examination. The appropriate number of hats, as specified below, shall be examined for conformance to the finished measurement requirements specified in 3.8. When a head size measurement deviates from a dimension and tolerance specified, the hat shall be penalized three points. When a measurement other than head size deviates from the dimension and tolerance specified, the hat shall be penalized one point. The lot shall be unacceptable if the total point value resulting from this examination exceeds the maximum acceptable point value specified below. Each size of hat present in the lot shall be represented in the sample selected for this examination.

Sampling provisions for dimensional examination

Lot size	Sample size	Maximum acceptable point value
Up thru 500	8	2
501 thru 3200	13	3
3201 thru 35000	20	5
35,001 and up	32	7

4.3.5 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5.

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Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping or inadequate stapling Bulged or distorted container
Content	Number per container is more or less than required

4.3.6 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded with required straps as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial as specified (see 6.2).

5.1.1 Level A. Each hat shall be individually unit packed in an oval shaped hat box of the design and dimensions as shown in figure 3. The box and cover shall be fabricated from chipboard that is completely covered on the inside with white color paper and the outside with high gloss black color paper. The stays around the top edge of the cover and the bottom edge of the

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box shall be trimmed with gold or yellow color paper, with 1/8 to 3/16 inch of the gold or yellow color showing on each side of the finished box. The box and cover strips shall be wire stitched. The box shall be provided with a carrying cord made of two strands of gold color rayon yarn over a cotton core. The strands shall be twisted together to form a cord approximately 1/4 inch in diameter and 29 inches long. The cord shall be laced through brass eyelets on each side of the box in a manner to form a carrying handle on one side and to extend across the cover to the other side of the box in a manner that will prevent the cover from slipping off the box when it is carried (see figure 3). Each end of the cord shall be secured on the inside of the box by a small knot or by any other means which will permit the ends of the cord to be as flat as possible against the sides of the box so as not to interfere with the packaging of the hat. Each hat shall be placed within the box, positioned over a snug-fitting cylindrical paperboard support, using white tissue paper to support the crown, cover the top and to prevent shifting of the hat in any direction. The cylindrical paperboard support shall be 3 ± 1/16 inches in height with the ends overlapping approximately 1/2 inch and secured by taping, stapling, or other mechanical means. The hats shall be further unit packed by inserting each hat box into a fiberboard box conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 125 of PPP-B-636. Inside dimensions of each fiberboard box shall approximate 11-3/4 inches in length, 9-3/4 inches in width, and 5-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Closure of the fiberboard box shall be accomplished by taping the top and bottom flaps of the box with 2-inch wide gummed paper tape conforming to type III, grade B of PPP-T-45. The tape shall be centered over the seam formed by the closure of the outer flaps of the top and bottom and extend down over the end panels not less than 2 inches.

NOTE: A certificate of compliance shall be supplied and will be acceptable for the specified hat box requirements.

5.1.2 Commercial. Hats shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B or Commercial as specified (see 6.2).

5.2.1 Level A packing. Twenty-four hats of one size only, preserved as specified in 5.1, shall be packed in a snug-fitting fiberboard box conforming to style RSC, grade V2s of PPP-B-636. Level A unit packs shall be packed three in length, two in width, and four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 30 inches in length, 24 inches in width, and 26 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be closed in accordance with method III, waterproofed in accordance with method V, and reinforced as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.5. Shipping containers shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified (see 6.2). Strapping shall be limited to nonmetallic strapping, except for type II, class F loads.

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5.2.2 Level B packing. Twenty-four hats of one size only, preserved as specified in 5.1, shall be packed flat in a snug-fitting fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Level A unit packs shall be packed three in length, two in width and four in depth within a shipping container. Inside dimensions of each shipping container shall approximate 30 inches in length, 24 inches in width, and 26 inches in depth. Approximate dimensions are furnished as a guide only. Each shipping container shall be closed in accordance with method II as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.5.

5.2.2.1 Weather-resistant fiberboard containers. When specified (see 6.2), the shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636, except that the inspection shall be in accordance with 4.3.5.

5.2.3 Commercial packing. Hats preserved as specified in 5.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), hats packed as specified in 5.2.2 and 5.2.3 shall be palletized on a 4-way entry pallet in accordance with load type Ia of MIL-STD-147. Pallet types shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Pallets shall be fabricated from wood groups I, II, III, or IV of MIL-STD-731. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be number 89 in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or order, unit packs, shipping containers and palletized unit loads shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

5.4.1 Labels, hat box. Each hat box shall have affixed thereto a removable type paper size label backed with pressure-sensitive adhesive. The paper label shall be oval shaped, 2 inches in length by 1-1/2 inches in width, with a white background and black numerals 1 inch in height. The label shall be positioned on center at the narrow end of the box body.

6. NOTES

6.1 Intended use. The hats are intended for wear by female personnel of the Department of the Army, as a component of the woman's Army Green service uniform.

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6.2 Ordering data. Acquistion documents should specify the following:

a. Title, number, and date of this document.
b. When a first article is required (see 3.2, 4.2, and 6.4).
c. Size required (see 1.2).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. Type and class of unit load required (see 5.2.1).
f. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
g. When palletization is required (see 5.3).

6.3 Samples. For access to hat samples and shade samples, address the procuring office issuing the invitation for bids.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition instruments regarding arrangements for selection, inspection, and approval of the first article.

6.5 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirement of the document (see 3.3).

6.6 Pressing cloth. A silk organza has been successfully used as a pressing cloth.

6.7 Subject term (key word) listing.

Hat, service woman's
Polyester and wool
Wool

6.8 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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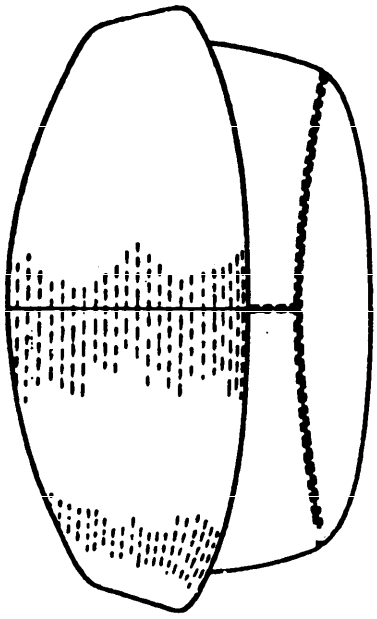
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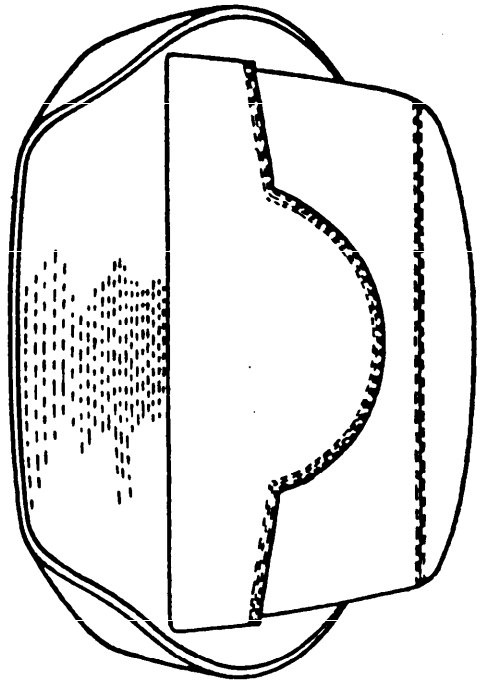
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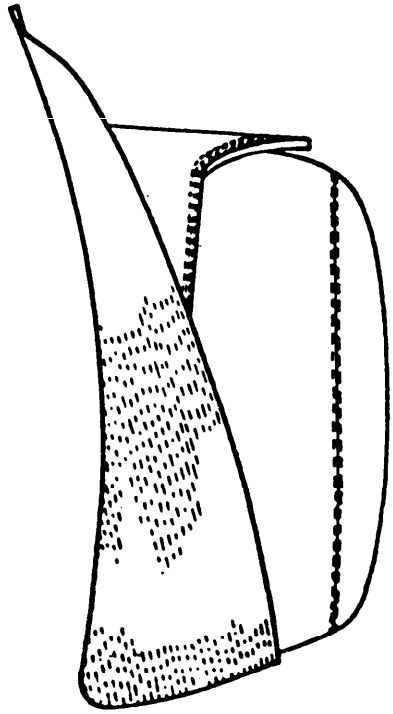
FIG. 1 HAT, SERVICE, WOMAN'S, WOOL, ARMY GREEN 344



BACK



FRONT



SIDE

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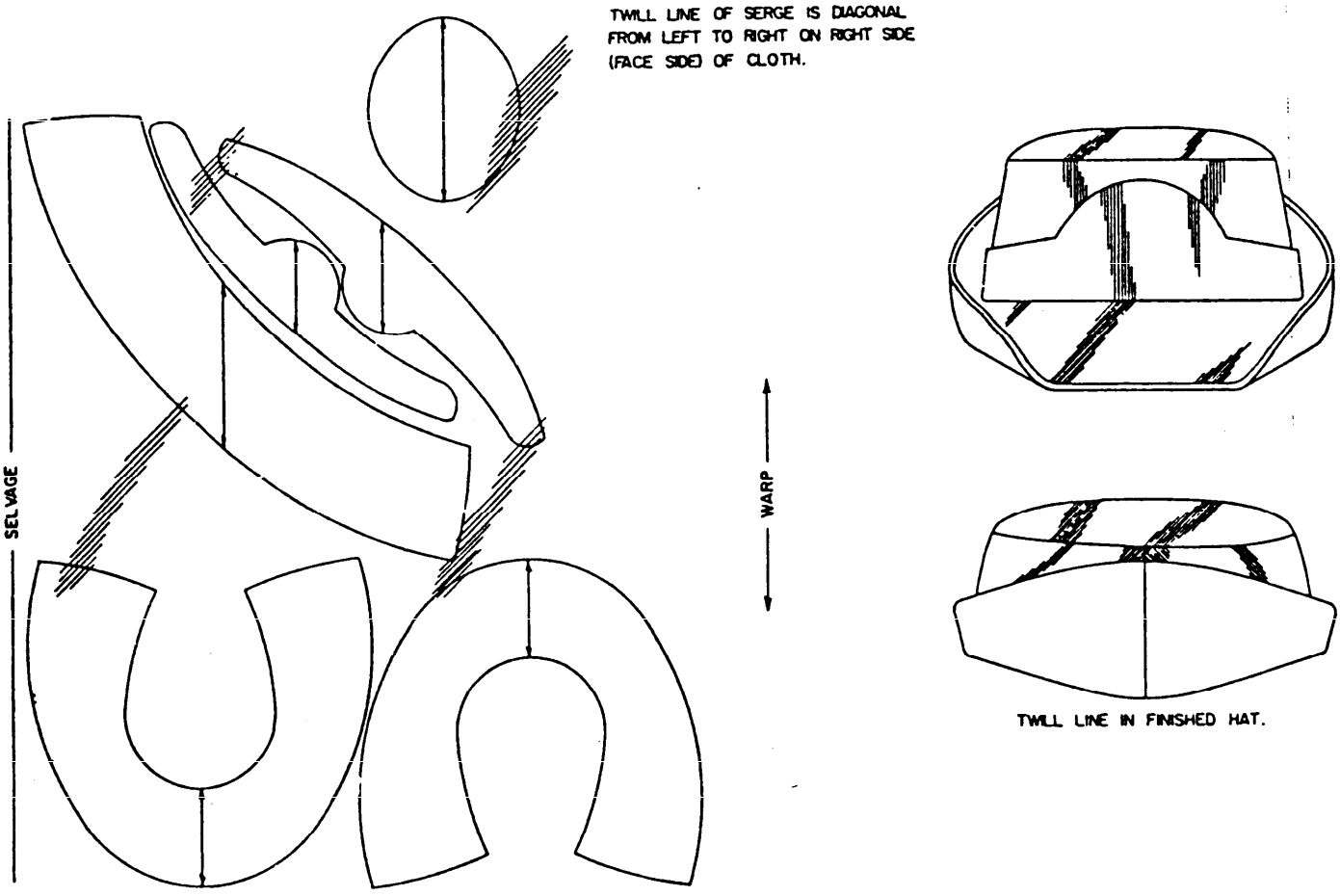


FIGURE 2 HAT, SERVICE, WOMAN'S WOOL, ARMY GREEN 44

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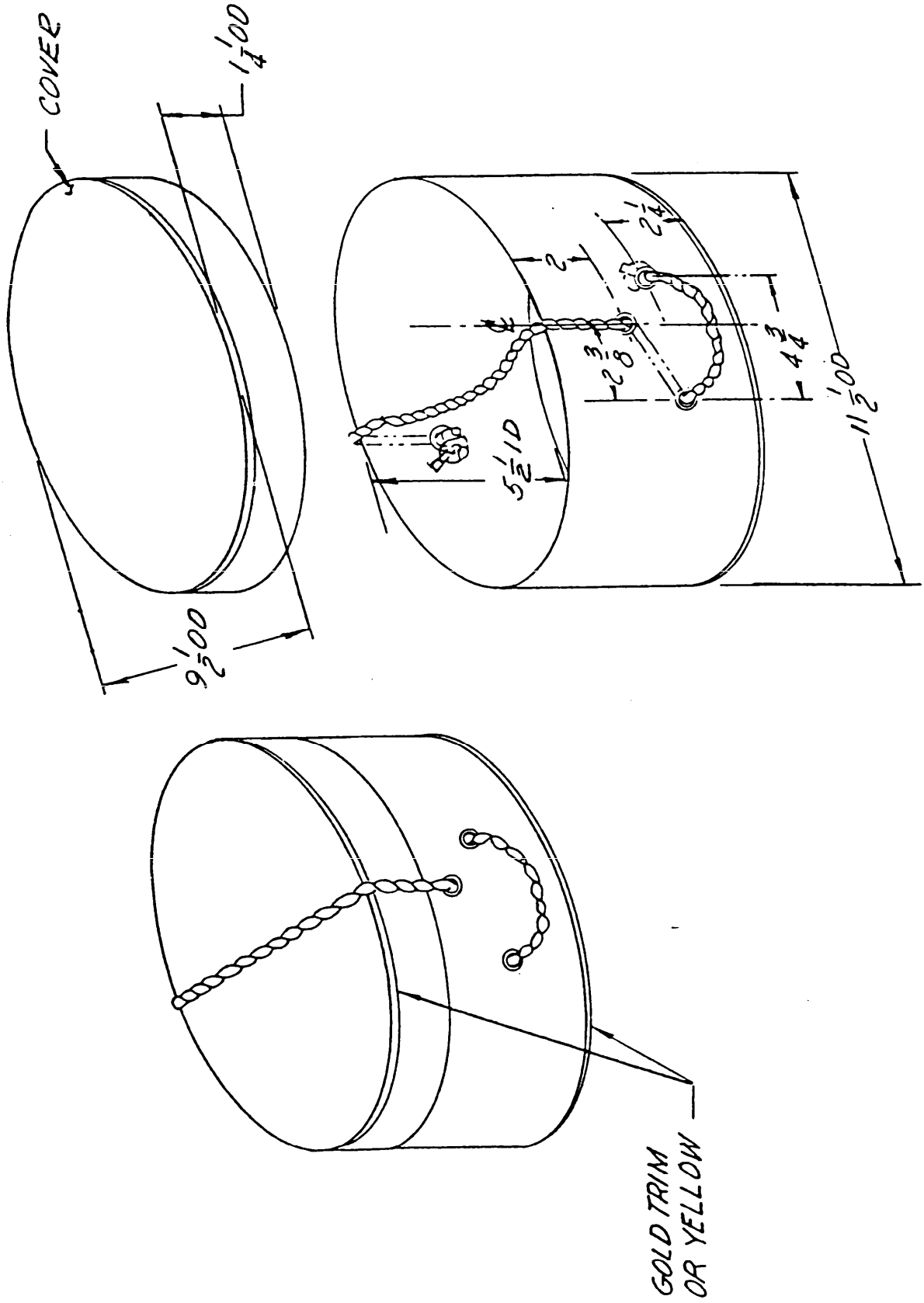


FIGURE 3. HAT BOX; HAT, SERVICE, WOMAN'S, WOOL OR POLYESTER AND WOOL.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-H-43162F(GL)	Hat, Service, Woman's, Wool or Polyester and Wool

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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